

Original LCTRSs

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{int} \times \text{int} \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{int} \Rightarrow \text{Process} \\ \text{sem} : \text{int} \times \text{int} \times \text{int} \Rightarrow \text{Semaphore} \\ \text{rdr7} : \text{Fcall} \\ \text{rdr9} : \text{int} \Rightarrow \text{Fcall} \\ \text{rdr10} : \text{int} \Rightarrow \text{Fcall} \\ \text{rdr11} : \text{int} \Rightarrow \text{Fcall} \\ \text{wrtr18} : \text{Fcall} \\ \text{wrtr19} : \text{Fcall} \\ \text{wrtr20} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1 :	cnfg(p(rdr7, 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr9(0), 0), p2, p3, sem(s, d, t), x, 0)	
2 :	cnfg(p(rdr9(y), 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr10(y), 0), p2, p3, sem(s1, d, t), x, 0)	[¬s = 0 ∧ s1 = s − 1]
3 :	cnfg(p(rdr9(y), 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr9(y), t), p2, p3, sem(s, d, t2), x, 0)	[s = 0 ∧ t2 = t + 2]
4 :	cnfg(p(rdr9(y), n), p2, p3, sem(s, d, t), x, 1) → cnfg(p(rdr10(y), 0), p2, p3, sem(s, d, t), x, 0)	[n = d ∧ ¬n = 0]
5 :	cnfg(p(rdr10(y), 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr11(x), 0), p2, p3, sem(s, d, t), x, 0)	
6 :	cnfg(p(rdr11(y), 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr7, 0), p2, p3, sem(s, d2, t), x, 1)	[¬t = d + 2 ∧ d2 = d + 2]
7 :	cnfg(p(rdr11(y), 0), p2, p3, sem(s, d, t), x, 0) → cnfg(p(rdr7, 0), p2, p3, sem(s1, d2, t), x, 0)	[t = d + 2 ∧ s1 = s + 1]
8 :	cnfg(p1, p(rdr7, 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr9(0), 0), p3, sem(s, d, t), x, 0)	
9 :	cnfg(p1, p(rdr9(y), 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr10(y), 0), p3, sem(s1, d, t), x, 0)	[¬s = 0 ∧ s1 = s − 1]
10 :	cnfg(p1, p(rdr9(y), 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr9(y), t), p3, sem(s, d, t2), x, 0)	[s = 0 ∧ t2 = t + 2]
11 :	cnfg(p1, p(rdr9(y), n), p3, sem(s, d, t), x, 1) → cnfg(p1, p(rdr10(y), 0), p3, sem(s, d, t), x, 0)	[n = d ∧ ¬n = 0]
12 :	cnfg(p1, p(rdr10(y), 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr11(x), 0), p3, sem(s, d, t), x, 0)	
13 :	cnfg(p1, p(rdr11(y), 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr7, 0), p3, sem(s, d2, t), x, 1)	[¬t = d + 2 ∧ d2 = d + 2]
14 :	cnfg(p1, p(rdr11(y), 0), p3, sem(s, d, t), x, 0) → cnfg(p1, p(rdr7, 0), p3, sem(s1, d2, t), x, 0)	[t = d + 2 ∧ s1 = s + 1]
15 :	cnfg(p1, p2, p(wrtr18, 0), sem(s, d, t), x, 0) → cnfg(p1, p2, p(wrtr19, 0), sem(s1, d, t), x, 0)	[¬s = 0 ∧ s1 = s − 1]
16 :	cnfg(p1, p2, p(wrtr18, 0), sem(s, d, t), x, 0) → cnfg(p1, p2, p(wrtr18, t), sem(s, d, t2), x, 0)	[s = 0 ∧ t2 = t + 2]
17 :	cnfg(p1, p2, p(wrtr18, n), sem(s, d, t), x, 1) → cnfg(p1, p2, p(wrtr19, 0), sem(s, d, t), x, 0)	[n = d ∧ ¬n = 0]
18 :	cnfg(p1, p2, p(wrtr19, 0), sem(s, d, t), x, 0) → cnfg(p1, p2, p(wrtr20, 0), sem(s, d, t), 1, 0)	
19 :	cnfg(p1, p2, p(wrtr20, 0), sem(s, d, t), x, 0) → cnfg(p1, p2, p(wrtr18, 0), sem(s, d2, t), x, 1)	[¬t = d + 2 ∧ d2 = d + 2]
20 :	cnfg(p1, p2, p(wrtr20, 0), sem(s, d, t), x, 0) → cnfg(p1, p2, p(wrtr18, 0), sem(s1, d, t), x, 0)	[t = d + 2 ∧ s1 = s + 1]
21 :	cnfg(p1, p2, p3, sem(s, d, t), x, a) → success	
22 :	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
23 :	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
24 :	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
25 :	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
26 :	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
27 :	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
28 :	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
29 :	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	

APR Problem

$$\{ \text{30 : } \text{cnfg(p(rdr7, 0), p(rdr7, 0), p(wrtr18, 0), sem(2, d, t), 0, 0) \Rightarrow success} \quad [(d > 0) \wedge t = d + 2] \}$$

BV-LCTRS Version

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{cnfg} : \text{Process} \times \text{Process} \times \text{Process} \times \text{Semaphore} \times \text{bitvec}_{32} \times \text{bitvec}_1 \Rightarrow \text{Cnfg} \\ \text{p} : \text{Fcall} \times \text{bitvec}_3 \Rightarrow \text{Process} \\ \text{sem} : \text{bitvec}_2 \times \text{bitvec}_3 \times \text{bitvec}_3 \Rightarrow \text{Semaphore} \\ \text{rdr7} : \text{Fcall} \\ \text{rdr9} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{rdr10} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{rdr11} : \text{bitvec}_{32} \Rightarrow \text{Fcall} \\ \text{wrtr18} : \text{Fcall} \\ \text{wrtr19} : \text{Fcall} \\ \text{wrtr20} : \text{Fcall} \\ \text{success} : \text{Cnfg} \\ \text{error} : \text{Cnfg} \end{array} \right\}$$

1 :	cnfg(p(rdr7, #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr9(#x00000000), #b000), p2, p3, sem(s, d, t), x, #b0)	
2 :	cnfg(p(rdr9(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr10(y), #b000), p2, p3, sem(s1, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
3 :	cnfg(p(rdr9(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr9(y), t), p2, p3, sem(s, d, t2), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
4 :	cnfg(p(rdr9(y), n), p2, p3, sem(s, d, t), x, #b1) → cnfg(p(rdr10(y), #b000), p2, p3, sem(s, d, t), x, #b0)	[n = d ∧ ¬n = #b000]
5 :	cnfg(p(rdr10(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr11(x), #b000), p2, p3, sem(s, d, t), x, #b0)	
6 :	cnfg(p(rdr11(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr7, #b000), p2, p3, sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
7 :	cnfg(p(rdr11(y), #b000), p2, p3, sem(s, d, t), x, #b0) → cnfg(p(rdr7, #b000), p2, p3, sem(s1, d2, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
8 :	cnfg(p1, p(rdr7, #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr9(#x00000000), #b000), p3, sem(s, d, t), x, #b0)	
9 :	cnfg(p1, p(rdr9(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr10(y), #b000), p3, sem(s1, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
10 :	cnfg(p1, p(rdr9(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr9(y), t), p3, sem(s, d, t2), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
11 :	cnfg(p1, p(rdr9(y), n), p3, sem(s, d, t), x, #b1) → cnfg(p1, p(rdr10(y), #b000), p3, sem(s, d, t), x, #b0)	[n = d ∧ ¬n = #b000]
12 :	cnfg(p1, p(rdr10(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr11(x), #b000), p3, sem(s, d, t), x, #b0)	
13 :	cnfg(p1, p(rdr11(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr7, #b000), p3, sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
14 :	cnfg(p1, p(rdr11(y), #b000), p3, sem(s, d, t), x, #b0) → cnfg(p1, p(rdr7, #b000), p3, sem(s1, d2, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
15 :	cnfg(p1, p2, p(wrtr18, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr19, #b000), sem(s1, d, t), x, #b0)	[¬s = #b00 ∧ s1 = sbvsub#b01]
16 :	cnfg(p1, p2, p(wrtr18, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, t), sem(s, d, t2), x, #b0)	[s = #b00 ∧ t2 = tbvadd#b010]
17 :	cnfg(p1, p2, p(wrtr18, n), sem(s, d, t), x, #b1) → cnfg(p1, p2, p(wrtr19, #b000), sem(s, d, t), x, #b0)	[n = d ∧ ¬n = #b000]
18 :	cnfg(p1, p2, p(wrtr19, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), #x00000001, #b0)	
19 :	cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, #b000), sem(s, d2, t), x, #b1)	[¬t = dbvadd#b010 ∧ d2 = dbvadd#b010]
20 :	cnfg(p1, p2, p(wrtr20, #b000), sem(s, d, t), x, #b0) → cnfg(p1, p2, p(wrtr18, #b000), sem(s1, d, t), x, #b0)	[t = dbvadd#b010 ∧ s1 = sbvadd#b01]
21 :	cnfg(p1, p2, p3, sem(s, d, t), x, a) → success	
22 :	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
23 :	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
24 :	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
25 :	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr19, n3), sem(s, d, t), x, a) → error	
26 :	cnfg(p(rdr10(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
27 :	cnfg(p(rdr11(y), n1), p(rdr10(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
28 :	cnfg(p(rdr10(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	
29 :	cnfg(p(rdr11(y), n1), p(rdr11(y), n2), p(wrtr20, n3), sem(s, d, t), x, a) → error	

APR Problem

{ 30 : cnfg(p(rdr7, #b000), p(rdr7, #b000), p(wrtr18, #b000), sem(#b10, d, t), #x00000000, #b0) ⇒ success [(dbvugt#b000) ∧ t = dbvadd#b010] }